SOHAM ROY

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EDUCATION

Birla Institute of Technology and Sciences(BITS), Pilani
Bachelor of Computer Science Engineering: 6 CGPA — Expected May 2028
Relevant Coursework: DSA, DBMS, AI, Digital Design, Computer Architecture., OS, Compiler Construction, Computer Networks
Secured an All India Rank of **6860** in the IIT-JEE examination
Secured **98.32 %ile** in the JEE Mains examination

WORK EXPERIENCE

CyberArk & BITS Pilani(Under Dr. Ashutosh Bhatiya)
Research Assistant — October 2025 – January 2026
- Collaborated on a **Zero-Trust Architecture** framework for securing LLM-based agents.
- Designed policies for safe tool invocation and auditing to mitigate prompt- and tool-injection attacks and worked on a paper.

Remote(Under Dr. Suresh Kumar)
Junior Researcher — September 2025 – December 2025
- Assisted in numerical analysis of cosmic background data; wrote scripts for **data preprocessing** and **spectral visualization**.
- Reviewed various research papers to understand **Cosmic Background Radiation.**

FIDE(Beckn Protocol) (building decentralized and interoperable P2P networks.)[GitHub]
Software Engineering Intern(Open Source Software) — May 2025 – July 2025
- Containerized **MCP**-based **microservices** on various API endpoints (Node.js, FastAPI) for **P2P** LLM trade orchestration.
- Optimized **RAG pipeline** reducing token usage by **32%**.

BITS Pilani(University-Under Dr. Radha Raman Mishra)
Research Assistant — November 2023 – May 2024
- Simulated and tested **3D-printed magnetic PLA** structures under **microwave exposure** for **adaptive space components**.
- Used **COMSOL Multiphysics** for **modeling and correlation** with lab data.

PROJECTS

FinOps Cloud Cost Forecaster [GitHub][Streamlit] — Aug 2026– Sep 2026
- Built and deployed an **ML dashboard** which takes 1 year data of cloud costs as input, runs it through **Meta Prophet** and gives a **forecasted computing** cost of up to **90 days** in the future.
- The clean **dashboard graphs** showcasing **time-series analysis** also gives vital information on **financial waste/excess** based on various **input factors** that the user can modify.

Q-Learning in a 3D Gridworld-Reinforcement Learning [GitHub] — Oct 2025 – Nov 2025
- Played around with **reinforcement learning,** specifically the **Q-Learning formula** and its effect on a **Hidden Markov Model** to generate **graphs** and **heat-maps** using **Matplotlib** with different test cases.
- A version of this project was also further implemented to an extent used in an **SCP based VR game** made by our team at the **university coding club** for the **enemy AI logic**.

AI-Agent for Unstructured Data Analysis(Deep Researcher) [GitHub] — Jan 2025 – May 2025
- Developed an AI assistant integrating **web scraping**, **PDF parsing** and **semantic search** to answer queries based on user uploaded research papers using **RAG** via **Langchain** and **ChromaDB**
- Benchmarked on **customized and BERT-based rubric**, achieving 90% similarity with top commercial LLMs (GPT-3.5, Llama 2).

P.T.O

Lexical Analyzer and Parser for a custom compiler [GitHub] Jan 2025 – May 2025

- Designed and implemented a **DFA** for a **lexical analyzer** and **parser** in C based on requirements and test cases provided by a professor.
- Uses a **twin-buffer I/O** to skip repeated file reads and builds a **LL(1) parsing table** for error and sync issues.

SKILLS

Languages and Databases: C++, C#, Java, Python, JavaScript, TypeScript, CSS, SQL, PostgreSQL, Verilog, Bash, MongoDB.
AI and Backend: Pytorch, TensorFlow, NodeJS, React, Langchain, ChromaDB, FAISS, Docker, Kubernetes, RabbitMQ, MLFlow
Developer Tools and Systems: Git, GitHub, COMSOL, Flutter, Grafana, NetCat, WireShark, Postman.
Certifications & Training: Cisco: Intro to Cybersecurity, Cisco: Ethical Hacking, German A1 Grade Exam: 51/60
Languages: English(Fluent), Hindi(Fluent), Marathi(Fluent), Bengali(mother-tongue), German(Intermediate)

ACTIVITIES

BITS Coding Club: Game Development Vertical

Senior member January 2024 – Present

- Built a **VR Game** from scratch. I worked primarily on **Blender** and Unity, and I expertised in **physics, animations** and **game design**.

Matrix: Film & Literature Society

Vice President August 2025 – Present

- Co-led the club on campus through 3 festivals involving public book-fests and movie-screenings. Revenue from stalls generated was 25+ lakhs INR

BITS Martial Arts

Team Member - 2 time medalist August 2023 – May 2025

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